Global Fuel Prices & Taxation Analytics

End-to-end project documentation

Prepared from the user-provided global fuel prices dataset and project-description video. The deliverable implements the main requested workflow: collection/loading, cleaning, database storage and SQL, analysis, visualizations, dashboard, narrative story, performance checks, web integration and documentation.

Project component	Implementation
Data processing	Pandas cleaning + derived features
Storage/query	SQLite database, indexes, 3 analytical views
Visualization	Plotly interactive charts + Matplotlib EDA charts
Web integration	Flask responsive dashboard + data story
BI handoff	Clean row-level CSV and five summary extracts
Testing	Unit tests for schema, row count, database views and keys

1. Executive summary

The supplied dataset contains 27,468 rows and 10 columns. After type conversion and required-field validation, 27,468 rows remain. The file spans 2020-01-06 to 2026-04-06, covers 84 countries and 7 regions, and contains 0 missing cells after cleaning. There were 0 exact duplicate rows in the original upload.

The project provides an interactive dashboard with filters for region, income level, subsidy level, country and date range; KPI cards; trend, regional, income, tax-price, correlation and country-ranking charts; and a five-scene data story. SQL and Tableau-ready extracts are provided for additional analysis.

2. Problem statement and audiences

The objective is to explore how petrol, diesel and LPG prices vary over time and across regions, income groups and subsidy categories, and how these patterns coexist with tax percentage and Brent crude values. The dashboard supports three illustrative audiences from the project brief: energy policy consultants comparing tax/subsidy patterns, supply-chain managers monitoring fuel cost variation, and energy-economics researchers examining group-level associations.

3. Data dictionary

Field	Type / meaning
date	Date of observation (converted to ISO date)
country	Country name
region	Global region category
income_level	High / Middle / Low income category
subsidy_level	Low / Medium / High / Very High category
petrol_usd_liter	Petrol price in USD per litre, as supplied
diesel_usd_liter	Diesel price in USD per litre, as supplied
lpg_usd_liter	LPG price in USD per litre, as supplied
brent_crude_usd	Brent crude value; unit/definition not supplied
tax_percentage	Tax percentage as supplied
year, month, quarter	Calendar features derived from date
avg_fuel_price_usd_liter	Mean of petrol, diesel and LPG price columns
fuel_price_spread_petrol_diesel	Petrol minus diesel price
fuel_price_spread_petrol_lpg	Petrol minus LPG price

4. Data preparation and quality

Steps: parse date values into datetime; coerce numeric measures to numeric types; remove rows missing required dimensions or numeric measures; remove exact duplicate rows; sort by country and date; derive year/month/quarter and price-spread/average fields; export the cleaned file; load all prepared records into SQLite. The uploaded file had no missing values and no exact duplicate rows, so the validation stage retained all 27,468 records. No extreme values were silently removed.

Quality caveat: observed minimum fuel prices of $0.01/litre and tax values of 0% and 65% were retained as source values. These may be valid, edge-case, or synthetic values; they should be checked against a data dictionary/source before formal decisions.

5. Database and SQL

SQLite table `fuel_prices` stores row-level records and derived columns. Indexes are created on date, country, region, income level and subsidy level to support common filters. Three views are included: `v_regional_fuel_summary`, `v_yearly_fuel_trends`, and `v_country_fuel_summary`. Additional reusable queries are in `sql/analysis_queries.sql`.

```
Example: SELECT region, AVG(petrol_usd_liter) AS avg_petrol FROM fuel_prices GROUP BY region
ORDER BY avg_petrol DESC;
```

6. Dashboard and visual design

The Flask app uses parameterized SQL filters and Plotly for interactive charts. Responsive CSS adapts filters and chart layout for smaller screens. KPI cards report filtered records, countries represented, average petrol price and average tax. Charts include: (1) fuel prices by year, (2) average petrol price by region, (3) petrol price by income group, (4) tax vs petrol scatter, (5) numeric correlation matrix, and (6) countries with highest average petrol prices. `/api/summary` exposes a small JSON summary and `/health` reports application/database health.

7. Five-scene data story

Scene 1 — Global overview

Summarizes selected record count and mean prices for petrol, diesel and LPG.

Scene 2 — Regional price gaps

Contrasts the highest and lowest regional petrol-price averages in the current filter.

Scene 3 — Time trend

Shows the minimum and maximum annual average petrol-price values in the filtered data.

Scene 4 — Income groups

Compares average petrol prices across income categories.

Scene 5 — Subsidies and policy context

Compares observed price averages by subsidy label, with a non-causal interpretation warning.

8. Exploratory findings (whole dataset)

- Average prices across all records: petrol $2.282/litre, diesel $2.134/litre and LPG $1.711/litre.
- Regional petrol average is highest in Europe ($3.702/litre) and lowest in Middle East ($1.226/litre), a gap of $2.476/litre.
- Average petrol price is $1.485/litre in 2020 and $2.647/litre in the partial 2026 data. Because 2026 only runs through April 6, it is not a full-year comparison.
- Across all rows, the mean tax percentage is 26.77%. Correlation is descriptive only and should not be interpreted as a causal tax effect.
- The supplied data shows 327 observations per country; this regularity is useful for comparison but does not independently verify the source's real-world provenance.

9. Performance testing and validation

The project includes integrity tests for CSV schema/row count/missing values, SQLite row count and views, and duplicate country-date keys. SQLite indexes are created on frequently filtered dimensions. Scatter points are capped at 5,000 in the dashboard to keep browser rendering responsive. To run checks: `python -m unittest discover -s tests -v`. For a formal benchmark, record page-load time and query latency on the target machine and document CPU/RAM/database size.

10. Running and integration

Install dependencies with `pip install -r requirements.txt`, then run `python app.py` and open `http://127.0.0.1:5000`. Dashboard: `/`; story: `/story`; health: `/health`; summary JSON: `/api/summary`. For Tableau, import `data/fuel_prices_cleaned.csv` or one of the aggregate CSVs. A Tableau-authored workbook is not included because Tableau Desktop/Public is a separate GUI application; the data extracts and build instructions are included.

11. Limitations and responsible interpretation

The dataset does not include source citations, data-generation notes, currency conversion details, a tax definition, or a precise unit for the Brent crude field. Results are descriptive; differences by income/subsidy category may be confounded by country, region, exchange rates, policy, and other factors. Validate data lineage and definitions before external publication, financial decisions or policy recommendations.

12. Deliverable inventory

- README.md — setup and component guide
- app.py + templates/ — responsive Flask dashboard and story
- data/fuel_prices_cleaned.csv — cleaned row-level dataset
- data/fuel_prices.db — SQLite database with indexes and views
- data/*_summary.csv — Tableau/Excel-ready aggregates
- sql/analysis_queries.sql — example SQL operations
- analysis.py — reproducible summary analysis
- outputs/charts/*.png — static exploratory visuals
- outputs/analysis_summary.json and data_quality.json — metrics
- tests/test_project.py — integrity tests
- docs/Project_Documentation.pdf — this document